

Unit 7: Sequences & Series

Honors Precalculus

Student

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7.1 Sequences

Objectives

- Define a sequence and identify terms and positions using subscript notation.
- Recognize patterns in sequences and write explicit formulas for the n th term.
- Write terms of a sequence defined recursively.
- Evaluate expressions involving factorial notation.
- Use summation (sigma) notation to write and evaluate sums.

Sequences

Sequences are written using subscript notation, as shown:

$$a_1, a_2, a_3, a_4, \dots, a_n, \dots$$

The numbers in the sequence are the _____.

The subscripts are the _____ of the terms.

Infinite Sequences

A _____ sequence is one where the terms get closer and closer to a single number as the sequence goes on.

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$$

A _____ sequence is one where the terms do not approach a single number.

$$2, 3, 5, 7, 11, \dots$$

$$1, -1, 1, -1, 1, -1, \dots$$

Honors: $\{a_n\}$ converges to L if for every $\varepsilon > 0$ there is an N with $|a_n - L| < \varepsilon$ whenever $n \geq N$.

Example 1

Write the first three terms of the sequence given by

$$a_n = \frac{(-1)^n}{2n-1}.$$

Example 2

Write an expression for the n th term (a_n) of the sequence

1, 3, 5, 7, ...

Example 3

Write an expression for the n th term (a_n) of the sequence

2, 5, 10, 17, ...

Recursive Sequences

Each term in a recursive sequence is defined by the term(s) before it.

A famous recursive sequence (Fibonacci):

0, 1, 1, 2, 3, 5, 8, 13, ...

Honors: Binet's formula gives a closed form: $F_n = \frac{1}{\sqrt{5}}[\varphi^n - (1 - \varphi)^n]$, where $\varphi = \frac{1+\sqrt{5}}{2}$.

Example 4

Write the first four terms of the sequence defined recursively as

$$a_1 = 3, \quad a_k = 2a_{k-1} + 1, \quad k \geq 2.$$

Factorial

If n is a positive integer, then n factorial is defined as

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdots 3 \cdot 2 \cdot 1.$$

As a special case, zero factorial is defined as

$$0! = 1.$$

Example 5–6

Simplify each factorial expression.

5) $\frac{8!}{2!6!}$

6) $\frac{n!}{(n-1)!}$

Summation Notation

$$\sum_{k=1}^n f(k)$$

$$\sum_{k=1}^n f(k) = f(1) + f(2) + f(3) + \cdots + f(n)$$

k is the *index*, 1 is the *lower limit*, n is the *upper limit*, and $f(k)$ is the *summand*.

Example 7

Evaluate

$$\sum_{k=3}^6 (1 + k^2).$$

Example 8

Evaluate

$$\sum_{n=0}^5 \frac{1}{n!}.$$

Example 9

Find the 3rd partial sum.

$$\sum_{k=1}^{\infty} \frac{3}{10^k}$$

Honors: Summation FormulasFor any positive integer n :

$$\sum_{i=1}^n c = cn \qquad \sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$$

These can be proved by mathematical induction.

Homework: Introduction to Sequences CIRCUIT worksheet.**7.2 Arithmetic Sequences and Series****Objectives**

- Recognize arithmetic sequences and identify the common difference.
- Write an explicit formula for the n th term of an arithmetic sequence.
- Insert arithmetic means between two given terms.
- Find finite partial sums of arithmetic series.
- Use arithmetic sequences and series to model real-life situations.

Arithmetic Sequences

An _____ is one that has a common difference, d . In other words, you _____ d to get to the next term.

1, 2, 3, 4, 5, 6, ... 0, 3, 6, 9, 12, ... 10, 5, 0, -5, -10, ...

An arithmetic sequence is like a _____ function.

The common difference is the _____.

n	a_n
1	a_1
2	a_2
3	a_3
n	a_n

Arithmetic Sequence Formulas

Arithmetic Sequences	Geometric Sequences
n th term:	n th term:
sum:	sum:
	infinite sum:

Honors: Generate or prove the n th-term formula by writing $a_n = a_1 + d + d + \cdots + d$ with $(n - 1)$ copies of d .

Example 1

Find a formula for the n th term of the arithmetic sequence whose common difference is -11 and whose first term is 66.

Example 2

The 4th term of an arithmetic sequence is 24 and the 13th term is 87. Write the first five terms of the sequence.

Arithmetic Means

Arithmetic means are the numbers that fit _____ two given terms in an arithmetic sequence, creating a larger sequence with a constant _____.

For example, two arithmetic means between 5 and 17 are:

$$5, \underline{9}, \underline{13}, 17$$

Honors: They are called “means” because they are the averages. For instance, the average of 5 and 13 is 9, and the average of 9 and 17 is 13.

Example 3

Insert three arithmetic means between 21 and 45.

The Sum of a Finite Arithmetic Series

Find the sum of the integers from 1 to 100.

This motivates the sum formula: $S_n = \frac{n}{2}(a_1 + a_n)$.

Example 4

Find the 101st partial sum of the arithmetic sequence 7, 9.5, 12, ...

Example 5

Express in terms of summation notation:

$$\frac{1}{4} + \frac{2}{9} + \frac{3}{14} + \frac{4}{19} + \frac{5}{24} + \frac{6}{29}$$

Example 6

A small business sells \$20,000 worth of sports memorabilia during its first year. The owner of the business has set a goal of increasing annual sales by \$15,000 each year for 19 years. Assuming that this goal is met, find the total sales during the first 20 years this business is in operation.

Homework: Arithmetic Sequences and Series CIRCUIT worksheet.

7.3 Geometric Sequences and Series

Objectives

- Recognize geometric sequences and identify the common ratio.
- Write an explicit formula for the n th term of a geometric sequence.
- Find geometric means between two terms.
- Find the sum of a finite geometric series.
- Determine when an infinite geometric series converges and find its sum.

Geometric Sequences

A _____ is one that has a common ratio, r . In other words, you _____ to get to the next term.

2, 10, 50, 250, ...

3, -6, 12, -24, ...

1, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, ...

Geometric Sequence Formulas

Arithmetic Sequences	Geometric Sequences
n th term: $a_n = a_1 + (n - 1)d$	n th term:
sum: $S_n = \frac{n}{2}(a_1 + a_n)$ $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$	sum:
	infinite sum:

Example 1

Find the 15th term of the geometric sequence whose first term is 20 and whose common ratio is 1.05.

Example 2

Write an equation for the n th term of the geometric sequence where $a_6 = 4$ and $r = 3$.

Geometric Mean

The _____ of two positive numbers a and b is $\sqrt{ab}$.

Geometric means between two terms of a geometric sequence are the missing terms that maintain a constant ratio.

Honors: Geometric means are always positive. Consider the geometric sequence 2, 6, 18, 54, 162 — the geometric mean of 2 and 18 is $\sqrt{2 \cdot 18} = 6$.

Example 3

Find three geometric means between 3.12 and 49.92.

Sum of a Finite Geometric Series

The sum of the first n terms of a geometric sequence with first term a_1 and common ratio $r \neq 1$ is

$$S_n = a_1 \left(\frac{1 - r^n}{1 - r} \right)$$

Honors: Derive by writing $S_n = a_1 + a_1r + \cdots + a_1r^{n-1}$, then $rS_n = a_1r + \cdots + a_1r^n$, and subtracting.

Example 4

Consider the sequence 8, 4, 2, ... Find the exact sum of the first twelve terms.

Example 5

A geometric series has 1 as its first term and 3 as its common ratio. The sum of the first n terms is 29,524. Find n .

Example 6

Find the sum: $\sum_{n=1}^{12} 4(0.3)^n$

Infinite Geometric Series

If $|r| < 1$, the infinite geometric series converges and its sum is

$$S = \frac{a_1}{1 - r}$$

If $|r| \geq 1$, the series _____ (no finite sum).

Example 7

Find the sum of $3 - \frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \cdots$, if it exists.

Example 8

Find the sum of the infinite geometric series $\sum_{n=1}^{\infty} 3\left(\frac{2}{3}\right)^n$

Homework: Geometric Sequences and Series CIRCUIT worksheet.

7.4 The Binomial Theorem

Objectives

- Identify patterns in binomial expansions.
- Use Pascal's Triangle to find binomial coefficients.
- Use the combination formula to compute $\binom{n}{r}$.
- Apply the Binomial Theorem to expand $(a + b)^n$.
- Find a specific term in a binomial expansion.

$$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$
$$(a+b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6ab^5 + b^6$$

- The coefficients increase and then decrease _____.
- The coefficient of the first and last term is _____.
- The exponents of a decrease while the exponents of b increase.
- Each expansion of $(a + b)^n$ has _____ terms.

Row 0:				1			
Row 1:			1		1		
Row 2:			1	2		1	
Row 3:		1	3		3		1
Row 4:		1	4	6		4	1
Row 5:	1		5	10		10	5
Row 6:	1	5	10	10	5	1	

In Row 4 the coefficients are 1, 4, 6, 4, 1 — these are the coefficients of $(a + b)^4$.

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Example 1

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + \binom{n}{n}b^n$$
$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

The $(r + 1)$ th term is:

$$\binom{n}{r} a^{n-r} b^r$$

Example 2

Use the Binomial Theorem to write the expansion of $(x - 2y)^5$.

Example 3

Write the expansion of $(3x + 1)^5$.

Example 4

Find the **sixth** term of $(x + 2y)^8$.

Example 5

Find the **fourth** term of $(3x - 5)^9$.

Example 6

Find the coefficient of the term x^7y^{12} in the expansion of $(2x - 3y^3)^{11}$.

Homework: Binomial Theorem CIRCUIT worksheet.

7.5 Counting Principles

Objectives

- Use the Fundamental Counting Principle to solve counting problems.
- Use permutations to count ordered arrangements.
- Use combinations to count unordered selections.
- Count arrangements of non-distinct (repeated) objects.

Fundamental Counting Principle

Let A and B be two events.

- The first event A can occur in _____ different ways.
- After A has occurred, B can occur in _____ different ways.

The number of ways that the two events can occur is _____.

Example 1

Airports around the world are universally known by a unique three-letter code. How many different three-letter airport codes are possible?

Example 2

Telephone numbers in an area code have 7 digits. How many different phone numbers are possible within each area code? Telephone numbers cannot begin with 0 or 1.

Permutations

A permutation is an arrangement of objects in a _____ where changing the sequence creates a different result.

The number of permutations of n elements taken r at a time:

$${}_nP_r = \frac{n!}{(n-r)!}$$

Example 3

Eight horses are running in a race. In how many ways can these horses come in 1st, 2nd, and 3rd?

Example 3 (continued)

What if you want to rank all 8 horses?

Combinations

A combination is a way of selecting items from a group where the order of selection _____.

The number of combinations of n elements taken r at a time:

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

Permutation	Combination
Arranging items on a shelf	Choosing pizza toppings
Ordering people in a line	Putting people on a committee
Phone numbers	Picking lottery numbers
Finishing positions in a race	Selecting cards from a deck

Example 4

A pizza delivery shop has 12 different toppings from which to choose. This week, if you buy a 2-topping pizza, you get 2 more toppings for free. How many ways can the special 4-topping pizza be made?

Example 5

A student has 16 different t-shirts and needs to pack exactly 5 t-shirts for a week-long camp. How many different combinations of t-shirts can be packed?

Example 6

A committee of 6 people is going to be created from 7 students and 5 teachers.

a) How many ways can the committee be created?

b) How many ways can the committee be created if exactly four members must be students?

Non-Distinct Permutations

Suppose a set of n objects has r_1 of one kind, r_2 of a second kind, r_3 of a third kind, and so on...

The number of permutations of n non-distinct objects is:

$$\frac{n!}{r_1! r_2! r_3! \cdots}$$

(Divide by the repeats.)

Example 7

How many distinguishable ways can the letters in TATTOOS be written?

Homework: Counting Principles CIRCUIT worksheet.

7.6 Probability

Objectives

- Define experiments, sample spaces, events, and probability.
- Distinguish between theoretical and experimental probability.
- Use the Addition Rule for mutually exclusive and inclusive events.
- Use the Multiplication Rule for independent and dependent events.
- Use permutations and combinations to compute probabilities.
- Use the complement to find probabilities.

Vocabulary

An _____ is any process of observation that leads to a single, well-defined outcome.

The _____ (S) is the set of all possible outcomes of an experiment.

- Rolling a six-sided die: $S = \{1, 2, 3, 4, 5, 6\}$
- Flipping two coins: $S = \{HH, HT, TH, TT\}$

An _____ (E) is any subset of the sample space — a collection of one or more outcomes.

_____ is a measure of how likely an event is to occur.

Probability

$$0 \leq P(E) \leq 1$$

Probability can be expressed as a _____, _____, or _____.

If $P(E) = 0$, then E is an _____ event.

If $P(E) = 1$, then E is a _____ event.

Theoretical vs. Experimental Probability

Theoretical Probability — based on _____.

- Assumes all outcomes are equally likely.
- Example: probability of rolling a 3 on a fair die is $\frac{1}{6}$.

Experimental Probability — based on _____.

- Example: flipping a coin 100 times and getting 47 heads gives $\frac{47}{100}$.

As the number of trials increases, experimental probability approaches theoretical probability.

Addition Rule

The Addition Rule calculates the probability that _____ of several events occurs.

Mutually Exclusive Events — events that _____ happen at the same time:

$$P(A \text{ or } B) = P(A) + P(B)$$

Inclusive Events — events that _____ happen at the same time:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Example 1

In a class of 30 students, 18 were born in the United States, 6 in China, 2 in Germany, 3 in India, and 1 in Brazil. What is the probability that a student was born in the United States or India?

Example 2

In a group of 27 students, 20 take Chemistry, 19 take History, 15 take both, and 3 take neither. What is the probability that a randomly chosen student studies Chemistry or History?

Multiplication Rule

The Multiplication Rule calculates the probability that _____ events all occur.

Independent Events — the occurrence of one event _____ the occurrence of the other:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Dependent Events — the outcome of the first event _____ the probability of subsequent events:

$$P(A \text{ and } B) = P(A) \cdot P(B | A)$$

Example 3

There are 9 brown boxes and 6 blue boxes on a shelf.

a) Ana chooses a box and *replaces* it. Sebastian does the same. What is the probability that they both choose a brown box?

b) Ana chooses a box and *keeps* it. Then Sebastian chooses a box. What is the probability that they both choose a brown box?

Example 4

In a recent survey, 85% of adults said they could *not* go a full day without their smartphone. If 6 adults are chosen at random, what is the probability that all 6 say they could not go a full day without their smartphone?

Probability with Permutations and Combinations

When outcomes involve selecting or arranging objects, use counting techniques:

$$P(E) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$$

Use **combinations** when order does not matter and **permutations** when order matters.

Example 5

A bag contains 8 red marbles and 6 blue marbles. If you randomly select 4 marbles, what is the probability that exactly 2 are red?

Example 6

A 4-digit PIN is selected. What is the probability that there are no repeated digits?

Example 7

A 4-digit PIN is created using digits 1–9 without repetition. What is the probability the PIN starts with an odd digit?

Complement

The _____ of an event E is the set of all outcomes in the sample space that are not in E .

If E is an event, then _____ is the complementary event.

$$P(E) + P(E') = 1 \quad \implies \quad P(E) = 1 - P(E')$$

Example 8

A jar contains 8 blue marbles and 4 yellow marbles. What is the probability of selecting a random group of 3 marbles that contains *at least one* yellow marble?

Example 9

Two six-sided dice are rolled. Find the probability that the sum of the numbers shown is greater than 3.

Homework: Probability CIRCUIT worksheet.

7.7 Conditional Probability

Objectives

- Understand and compute conditional probabilities.
- Find conditional probabilities from two-way tables.
- Find conditional probabilities from Venn diagrams.
- Use tree diagrams to solve conditional probability problems.

Conditional Probability

Conditional Probability represents how the _____ of one event changes when we know another event _____.

“ A given B ”:

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

Conditional Probability from a Two-Way Table**Example 1**

A high school surveyed 300 students about their primary mode of transportation to school.

	9th	10th	11th	12th	Total
Walk/Bike	26	19	14	11	70
Car	14	26	39	45	124
Bus	36	31	26	13	106
Total	76	76	79	69	300

a) What is the probability that a student is in the 12th grade and drives a car to school?

b) What is the probability that a student takes the bus, given that they are in 9th grade?

Example 2

People exiting a new ride at a theme park were asked whether they liked or disliked the ride.

	Child	Adult	Total
Liked	51	26	
Disliked	22	31	
Total			

Find the probability that a randomly selected person:

a) is an adult or disliked the ride

b) is an adult *given that* he or she disliked the ride

Conditional Probability from a Venn Diagram

Example 3

In a class of 25 students, 14 use Instagram and 16 use TikTok. One student uses neither platform and six students use both platforms. One student is randomly selected from the class. What is the probability that the student uses Instagram given that he or she uses TikTok?

Example 4

In a class of 40 students, 34 like bananas, 22 like pineapples, and 2 dislike both fruits. If a student is randomly selected, find the probability that the student:

- a) likes bananas given that he or she likes pineapples

- b) dislikes pineapples, given that he or she likes bananas

Conditional Probability from a Tree Diagram

Example 5

Emma studies in her bedroom 40% of the time and at the library 60% of the time. In her bedroom, she does math 70% of the time and English 30% of the time. At the library, she does math 50% of the time and English 50% of the time. If you know Emma is working on math, what is the probability she is studying in her bedroom?

Example 6

Basket A contains 3 red and 2 white tickets. Basket B contains 4 red and 1 white. A die with 4 faces marked A and 2 faces marked B is rolled and used to select basket A or B. A ticket is then selected from this basket. Determine the probability that the ticket was chosen from B, given it is red.

Homework: Conditional Probability CIRCUIT worksheet.